

Aditi Kulkarni

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SUMMARY

Senior software engineer with 8+ years building high-throughput backend systems and distributed data pipelines across API services. Experienced in owning end-to-end development of large-scale, production-ready features, spanning design, development, infra/observability, and safe rollouts.

TECHNICAL SKILLS

Languages: Python 3, Golang, Java, Scala, SQL

Data & Messaging: Kafka, Spark, ClickHouse, Redis, Elasticsearch, MongoDB, GCP BigQuery, GraphQL

Infra & Tooling: Kubernetes, Docker, AWS, Ansible, CI/CD, Prometheus, Grafana, Git, JIRA, Agile

WORK EXPERIENCE

Walmart

Sunnyvale, CA

Senior Software Engineer

Nov 2024 – Present

- **Driver Language Preference Enablement (Python):** Architected a Python NLP-powered translation pipeline delivering Spanish language support to 30% of Walmart's driver base, reducing onboarding friction for non-English speakers; built as modular, pytest-covered services
- Mentored developers across teams to accelerate parallel implementation of pipeline components
- **Real-Time Events Tracking & Incident Detection:** Built a PySpark/Scala streaming pipeline processing >3M Kubernetes events/day, persisting results to ClickHouse and Redis behind a GraphQL API with <3 ms query latency
- Cut incident mitigation time to <5 minutes via Prometheus and Grafana alerting
- **Local Rate Limiting:** Designed and rolled out per-service rate limiting across the microservice fleet; analyzed traffic patterns to set thresholds and instrumented dashboards, preventing 3 major production incidents from resource exhaustion and cascading failures
- **Shopper Fraud Reduction – Last Mile Delivery:** Built guardrails for shopper drop workflows with validation checks and pharmacy-order edge-case handling, gated behind feature flags for safe phased rollout
- Achieved a 95% reduction in orders stuck at pick-complete, directly improving customer satisfaction

Software Development Engineer III

Jun 2022 – Nov 2024

- **Distributed Deployment Automation (Python):** Built Python automation deploying plugins to 15,000+ applications; overcame single-core constraints via multi-pod parallelism on Kubernetes
- Reduced a 3-day, 4-engineer process to under 5 hours and shipped a self-serve Slack bot interface for team-wide deployments
- **Dependency Mapping & Remediation System:** Built an end-to-end Java/Golang service for production troubleshooting across 60,000+ applications, exposing northbound/southbound dependencies via GraphQL over MongoDB and Elasticsearch
- Reduced MTTD and MTTR from >6 hours to consistently under 1 hour

Red Hat Inc

Los Angeles, CA

Software Engineer

Jul 2019 – Jun 2022

- **Business Logic Automation – Home Mortgages Platform:** Led development of a Java 8/Maven application embedding 1,500+ business rules and JUnit tests, cutting loan processing time from 1 week to under 1 hour
- Partnered with SMEs to translate complex domain logic into automated workflows, reducing manual intervention to critical errors only

KPMG India

Data Analyst

Jun 2016 – Jul 2017

- Built customer personas and journey maps through exploratory data analysis in Python, identifying service gaps and driving a 40% improvement in operational efficiency

EDUCATION

University of Southern California

Los Angeles, CA

M.S. in Electrical and Computer Engineering

2017 – 2019

Birla Institute of Technology and Science (BITS)

Pilani, India

B.E. (Hons.) Electrical and Electronics

2012 – 2016

PUBLICATIONS

Kulkarni et al., "Robotic Assistance in Indoor Navigation for People Who are Blind," Carnegie Mellon University Robotics Institute, funded by NSF National Robotics Initiative (IIS-1317989) ([Link](#))